Mingsong Yan

Visiting Assistant Professor
Department of Mathematics
University of California, Santa Barbara

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Research Interests

My research focuses on the mathematical foundations of *efficient* and *interpretable* machine learning. Using tools from optimization, numerical analysis, probability theory, dynamical systems, and functional analysis, I study how mathematical structure shapes representation, approximation, convergence, stability, and transferability in modern learning models. On the efficiency side, I study sparse regularization, low-rank approximation, size transferability, and optimization mechanisms that can reduce computational cost and improve scalability. On the interpretability side, I study hypothesis spaces for deep learning and in-context learning with transformers, including connections to kernel methods and iterative algorithms.

Employment

University of California, Santa Barbara — Isla Vista, CA
Visiting Assistant Professor, Department of Mathematics — *July 2024 – Present*

Education

Old Dominion University — Norfolk, VA
Ph.D., Computational and Applied Mathematics — *Sep. 2018 – Aug. 2024*
Dissertation: Deep Learning in Reproducing Kernel Banach Spaces
Advisor: Prof. Yuesheng Xu

Jilin University — Changchun, China
B.S., Tang Aoqing Honors Program in Mathematics — *Sep. 2014 – July 2018*

Georgia Institute of Technology — Atlanta, GA
Undergraduate Exchange Program in Mathematics — *Jan. 2017 – May 2017*

Publications

◇: Alphabetical authorship

Refereed Journal Articles

1. **Mingsong Yan**, Charles Kulick, Sui Tang. *On the Convergence and Size Transferability of Continuous-Depth Graph Neural Networks*. Accepted for publication in *SIAM Journal on Mathematics of Data Science*. arXiv:2510.03923, 2025.
2. Rui Wang, Yuesheng Xu, **Mingsong Yan**◇. *Hypothesis Spaces for Deep Learning*. *Neural Networks*, 193, 2026.
3. Ronglong Fang, Yuesheng Xu, **Mingsong Yan**◇. *Inexact Fixed-Point Proximity Algorithm for the ℓ_0 Sparse Regularization Problem*. *Journal of Scientific Computing*, 100(2):58, 2024.
4. Rui Wang, Yuesheng Xu, **Mingsong Yan**◇. *Sparse Representer Theorems for Learning in Reproducing Kernel Banach Spaces*. *Journal of Machine Learning Research*, 25(93):1–45, 2024.
5. Qianru Liu, Rui Wang, Yuesheng Xu, **Mingsong Yan**◇. *Parameter Choices for Sparse Regularization with the ℓ_1 Norm*. *Inverse Problems*, 39(2):025004, 2023.

Refereed Conference Proceedings

6. Ziyue Liu, Ruijie Zhang, Zhengyang Wang, **Mingsong Yan**, Zi Yang, Paul Hovland, Bogdan Nicolae, Franck Cappello, Sui Tang, Zheng Zhang. *CoLA: Compute-efficient Pre-training of LLMs via Low-rank Activation*. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2025.

Preprints

7. **Mingsong Yan**, Zhida Wang, Sui Tang. *Zero-Shot Size Transfer for Neural ODEs on Sparse Random Graphs: Graphon Limits and Adjoint Convergence.* arXiv:2606.26662, 2026.
8. **Mingsong Yan**, Dongyang Li, Charles Kulick, Sui Tang. *Transformers Can Implement Preconditioned Richardson Iteration for In-Context Gaussian Kernel Regression.* arXiv:2605.08475, 2026.
9. Lixin Shen, Rui Wang, Yuesheng Xu, **Mingsong Yan**$^{\diamond}$. *Sparse Deep Learning Models with the ℓ_1 Regularization.* arXiv:2408.02801, 2024.

Presentations

Invited Talks

Joint Mathematics Meetings, Chicago, IL	Jan. 2027 (Upcoming)
SIAM Conference on Mathematics of Data Science, Salt Lake City, UT	Nov. 2026
International Conference on Computational Harmonic Analysis, Vanderbilt University, Nashville, TN	May 2026
Computational Analysis Seminar, Vanderbilt University, Nashville, TN	Apr. 2026
SIAM Conference on Uncertainty Quantification, Minneapolis, MN	Mar. 2026
Mathematical Congress of the Americas, Miami, FL	July 2025
VAP Colloquium, UC Santa Barbara, CA	Feb. 2025
CMSE Brownbag Seminar, Michigan State University (Virtual)	Nov. 2024
Applied Math/PDE/Data Science Seminar, UC Santa Barbara, CA	Oct. 2024
ODU Mathematics Awareness Conference, Norfolk, VA	Apr. 2024
ODU Mathematics Awareness Conference, Norfolk, VA	Apr. 2023

Contributed Talks

Southern California Applied Mathematics Symposium, Loyola Marymount University, Los Angeles, CA	Apr. 2026
Annual WiMSoCal Symposium, Loyola Marymount University, Los Angeles, CA	Feb. 2026
AMS Spring Eastern Sectional Meeting, Washington, DC	Apr. 2024
Joint Mathematics Meetings, San Francisco, CA	Jan. 2024

Posters

IPAM Workshop: Foundations of Interpretability, UCLA, CA	Sep. 2026
NeurIPS 2025 Workshop: New Perspectives in Graph Machine Learning, San Diego, CA	Dec. 2025
NSF CompMath Meeting, University of Utah, UT	May 2025

Research Visits

Invited Member, SLMath AxIOM: *Machine Learning for Mathematics*, Berkeley, CA	Mar. 2027 (Upcoming)
Visiting Scholar, Department of Mathematics, Sun Yat-sen University, Guangzhou, China	June 2019

Grants and Funding

AMS-Simons Travel Grant ($5,000)	July 2025 – June 2027
ICCHA26 Shanks Conference Travel Award	June 2026
AMS-NSF Travel Grant for Mathematical Congress of the Americas	July 2025
NSF CompMath Meeting Travel Award	May 2025
AMS Travel Grant for Joint Mathematics Meeting	Jan. 2024

Honors and Awards

Mochizuki Memorial Instructor Award, Department of Mathematics, UC Santa Barbara	June 2026
Phillip R. Wohl Memorial Scholarship, Old Dominion University	May 2023
Cheng Innovative Research Assistantship, Old Dominion University	Summer 2022
Dominion Graduate Scholar, Old Dominion University	2018 – 2020
National Scholarship (Undergraduate), Jilin University, China	2016

Teaching Experience

Instructor – University of California, Santa Barbara

Ordinary Differential Equations (Math 119A)	Fall 2026
Numerical Analysis (Math 104B)	Winter 2026
Introduction to Numerical Analysis (Math 104A)	Fall 2025
Differential Equations (Math 4B)	Spring 2025 **(152 students)**
Transition to Higher Mathematics (Math 8)	Fall 2024; Winter 2025; Spring 2026; Fall 2026
Calculus II (Math 3B)	Fall 2025 **(149 students)**
	Fall 2024 **(111 students)**

Instructor – Old Dominion University

Precalculus I (Math 162)	Fall 2022; Spring 2023; Summer 2023; Spring 2024
	Fall 2020; Summer 2021; Fall 2021; Spring 2022

Teaching Assistant – Old Dominion University

Computational Linear Algebra (Math 616)	Fall 2023
Ordinary Differential Equations (Math 307)	Fall 2023
Introductory Linear Algebra (Math 316)	Fall 2023

Mentoring Experience

University of California, Santa Barbara

- Dongyang Li and Charles Kulick (Graduate students) 2025 – 2026
  Co-mentored with Prof. Sui Tang.
  Project: *Transformers Can Implement Preconditioned Richardson Iteration for In-Context Gaussian Kernel Regression.*
- Zhida Wang (Undergraduate student), Fangqian Zhang (Graduate student) 2025 – 2026
  Co-mentored with Prof. Sui Tang.
  Project: *Zero-Shot Size Transfer for Neural ODEs on Sparse Random Graphs: Graphon Limits and Adjoint Convergence.*
- Charles Kulick (Graduate student) 2024 – 2025
  Co-mentored with Prof. Sui Tang.
  Project: *On the Convergence and Size Transferability of Continuous-Depth Graph Neural Networks.*

Professional Service

Organizer, Applied Math/PDE/Data Science Seminar, UC Santa Barbara	Sep. 2024 – Present
Teaching Assistant, Data Science Bootcamp, Old Dominion University	May 2023

Referee Service

NeurIPS 2026
Inverse Problems (co-review with Prof. Sui Tang)
Sampling Theory, Signal Processing, and Data Analysis
Journal of Nonlinear, Complex and Data Science
Scientific Reports